Exercise Sheet 1: Systems of Linear Equations

October 2024

1 Matrices and Gaussian Elimination

Solving linear systems using elimination method

Recall: Common method to solve a linear system using Gaussian/Gauss–Jordan elimination:

1) Construct a matrix (called the augmented matrix) corresponding to the system.

2) Using *elementary row operations* on matrices to reduce the augmented matrix to an upper triangular matrix.

3) Using back-substitution to solve the final system.

Exercise 1. Which of the following are linear equations in x_1, x_2 and x_3?

1. $x_1 + 5x_2 - \sqrt{x}_3 = 1$
2. $x_1^{-2} + x_2 + 8x_3 = 5$
3. $\pi x_1 - \sqrt{2}x_2 + \frac{1}{3}x_3 = 7^{1/3}$
4. $x_1 - 3x_2 + \frac{\sqrt{4-\sqrt{32}}}{\sqrt{5}}x_3 = 2$

Exercise 2. The following matrices are the augmented matrices of systems of linear equations in variables $x_1, x_2, \ldots$. Write down the corresponding systems explicitly and solve them.

1.
$$\begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 4 \end{bmatrix}$$

2.
$$\begin{bmatrix} 1 & 0 & 0 & 4 & -1 \\ 0 & 1 & 0 & 2 & 6 \\ 0 & 0 & 1 & 3 & 2 \end{bmatrix}$$

3.
$$\begin{bmatrix} 1 & 6 & 0 & 0 & 4 & -2 \\ 0 & 0 & 1 & 0 & 3 & 1 \\ 0 & 0 & 0 & 1 & 5 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

4.
$$\begin{bmatrix} 1 & 2 & 0 & 3 & 0 & 7 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

Exercise 3. Solve the following linear systems using Gaussian or Gauss–Jordan (back-substitution) elimination. Write clearly all elementary row operations you used.

1.
$$\begin{aligned} x + y + 2z &= 0 \\ 2x + 4y - 3z &= 1 \\ 3x + 6y - 5z &= 0. \end{aligned}$$

2.

$$\begin{aligned} 2x + 2y + 2z &= 0 \\ -2x + 5y + 2z &= 1 \\ 8x + y + 4z &= -1. \end{aligned}$$

3.

$$\begin{aligned} x + 2y - t + w &= 1 \\ 3y + z - w &= 2 \\ z + 7t &= 1. \end{aligned}$$

4.

$$\begin{aligned} x - 2y + z - 4t &= 1 \\ x + 3y + 7z + 2t &= 2 \\ x - 12y - 11z - 16t &= 5. \end{aligned}$$

5.

$$\begin{aligned} 2I_1 - I_2 + 3I_3 + 4I_4 &= 9 \\ I_1 - 2I_3 + 7I_4 &= 11 \\ 3I_1 - 3I_2 + I_3 + 5I_4 &= 8 \\ 2I_1 + I_2 + 4I_3 + 4I_4 &= 10. \end{aligned}$$

6.

$$\begin{aligned} 2u - v &= 0 \\ -u + 2v - w &= 0 \\ -v + 2w - z &= 0 \\ -w + 2z &= 5. \end{aligned}$$

Exercise 4. For which real values of a will the following system have no solutions? Exactly one solution? Infinitely many solutions?

$$\begin{aligned} x + 2y - 3z &= 4 \\ 3x - y + 5z &= 2 \\ 4x + y + (a^2 - 14)z &= a + 2. \end{aligned}$$

Exercise 5. Determine the real numbers b_1, b_2, b_3 such that the following system has at least one solution:

$$\begin{aligned} -x_1 + 2x_2 - 3x_3 + x_4 &= b_1 \\ 2x_1 - x_2 + 2x_3 - 3x_4 &= b_2 \\ 3x_1 + x_3 - 5x_4 &= b_3. \end{aligned}$$

Exercise 6. Solve the following system with respect to the real parameter m:

$$\begin{aligned} x + my + z &= m \\ 2x + (m-1)y + 3z &= 1 \\ mx + y + 3z &= 2m - 3. \end{aligned}$$